Math 74: Algebraic Topology

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1 Topology

Definition 1.1. *A topology on a set* X *is a collection* J *of subsets of* X *having the following properties.*

1. $\emptyset, X \in J$
2. *The union of the elements of any sub-collection of* J *is in* J.
3. *The intersection of finitely many elements in* J *is in* J.

A set for which a topology has been specified is said to be a topological space. We think of it as a pair (X, J).

Definition 1.2. *Elements of* J *are called open sets of the topological space* (X, J).

Definition 1.3. *A subspace* Y *of a topological space* (X, J) *is a subset* $Y \subset X$ *which is given the 'subspace topology' where the open sets of* $Y = \{U \cap Y : U \in J\}$

Definition 1.4. *Let* (X_1, J_1) *and* (X_2, J_2) *be topological spaces. A map* $f : (X_1, J_1) \to (X_2, J_2)$ *is said to continuous if* $f^{-1}(U) \in J_1$ *for all* $U \in J_2$.

Definition 1.5. *Let* (X, J) *and* (X', J') *be topological spaces. Then they are homeomorphic if* $\exists$ *a continuous bijection* $f : (X, J) \to (X', J')$. *So that* f^{-1} *is also continuous.*

Exercise: Show that any two open intervals of $\mathbb{R}$ are homeomorphic (where we put subspace topology on the open intervals).

Solution: Let (a, b) and (c, d) be open intervals. **TODO:**

Remark 1.6. *Number of path connected topologies is preserved by continuous maps. For instance,* $[0, 1]$ *is not homeomorphic to* $(0, 1)$, *and* $\mathbb{R}$ *is not homeomorphic to* $\mathbb{R}^2$.

Exercise: Is $\mathbb{R}^2$ homeomorphic to $\mathbb{R}^3$?

Exercise: Is $S^2 (= x^2 + y^2 + z^2 = 1)$ homeomorphic to $T^2 = S^1 \times S^1$?

Exercise: Is S^2 homeomorphic to $S^1 \times [a, b)$?

Definition 1.7. *If f and f' are continuous maps from spaces X to Y we say that f is homotopic to f' if $\exists$ a continuous map $F : X \times I = [0, 1] \to Y$ such that $F(x, 0) = f(x)$ and $F(x, 1) = f'(x)$. F is called the homotopy between f and f', and we write $f \simeq f'$.*

Definition 1.8. ***TODO:** Product Topology*

Definition 1.9. *Define a path f as $f : [0, 1] \to X$ with $f(0) = x_0$ and $f(1) = x$.*

Definition 1.10. *Two paths $f, f' : [0, 1] \to X$ are said to be path homotopic if there exist a continuous map $F : I \times I \to X$ such that $\forall s \in I$,*

$$F(s, 0) = f(s), \qquad F(s, 1) = f'(s)$$

and $\forall t \in I$,

$$F(0, t) = x_0, \qquad F(0, t) = x$$

Then we write $f \underset{p}{\simeq} f'$.

Example 2. *Let $f : X \to \mathbb{R}^2$ and $g : X \to \mathbb{R}^2$ then $f \simeq g$.*

Proof. Want to find F such that $F : X \times I \to \mathbb{R}^2$. $F(x, 0) = f(x), F(x, 1) = f'(x)$ for all $x \in X$. The following map (straight line homotopy) works.

$$F(x, t) = (1 - t)f(x) + tf'(x)$$

□

Lemma 2.1. *The relations $\simeq$ and $\simeq_p$ are equivalence relations.*

Proof.

1. Reflexive: $f \simeq f$ by $F(s, t) = f(s)$.
2. Symmetric: Let $f \simeq g$ by the homotopy $F(s, t)$. We can show $g \simeq f$ by $F(w, t) = F(s, 1 - t)$.
3. Transitive: Let $f \simeq g$ by $F(s_1, t)$ and $g \simeq h$ by $G(s_2, t)$. We can show that $f \simeq h$ by $H(s, t) = \begin{cases} s = s_1, \ 0 \leq t \leq 1/2 \\ s = s_2, \ 1/2 \leq t \leq 1 \end{cases}$ **TODO:** check and also prove for $\simeq_p$

Thus path-homotopies are equivalence relations and we write [f] to denote the path-homotopy equivalence class of f. □

Lemma 2.2 (Pasting Lemma)**.** *Let $X = A \cup B$ be a topological space with $A, B \subset X$ are closed. Let $f : A \to Y, g : B \to Y$ be continuous and $f(x) = g(x), \forall x \in A \cap B$, then*

$$h(x) \begin{cases} f(x), \ x \in A \\ g(x), \ x \in B \end{cases}$$

is continuous.

Definition 2.3. *Let $f : I \to X$ be a path from x_0 to x_1 and $g : I \to X$ be a path from x_1 to x_2, then we define a new path*

$$f \star g : I \to X$$

from x_0 to x_2 by

$$f \star g(t) = \begin{cases} f(2t), \ t \in [0, 1/2] \\ g(2t-1), \ t \in [1/2, 1] \end{cases}$$

$\star$ is composition of paths.

Some Properties of '$\star$'.

1. *'$\star$' is well-defined operation in the path homotopy equivalence class,*

$$[f] \star [g] = [f \star g]$$

2. *Identity:*

$$[f] \star [e_{x_1}] = [f], \quad [e_{x_0}] \star [f] = [f]$$

   where e_{x_t} is the constant path based at $x \in X$ and $t \in [0, 1]$

3. *Inverse:*

$$[f] \star [\bar{f}] = [e_{x_0}], \quad [\bar{f}] \star [f] = [e_{x_1}]$$

   where $\bar{f}(t) = f(1-t)$.

4. *Associative:*

$$[f] \star ([g] \star [h]) = ([f] \star [g]) \star [h]$$

Proof.

1. Let $f \simeq_p f'$ by F and $g \simeq g'$ by G, then we can show that $f \star g \simeq f' \star g'$ by defining $H : I \times I \to X$ such that

$$H(s, t) = \begin{cases} F(2s, t), \ s \in [0, 1/2] \\ G(2s-1, t), \ s \in [1/2, 1] \end{cases}$$

2. $f \star e_{x_1} \simeq f$ by $H : I \times I \to X$ such that

$$H(s,t) = \begin{cases} f\left(\frac{2\cdot s}{2-t}\right), \ s \in \left[0, \frac{2-t}{2}\right] \\ x_1, \ s \in \left[\frac{2-t}{2}, 1\right] \end{cases}$$

3. $f \star \overline{f} \simeq_p e_{x_0}$.

   Define a path $\alpha(t)$ that starts at x_0 and ends at $f(t)$. Then the path-homotopy is given by $\alpha(t) \star \overline{\alpha}(t)$.

$$H(s,t) = \begin{cases} f(2st), \ s \in [0, 1/2] \\ f(2t(1-s)), \ s \in [1/2, 1] \end{cases}$$

   TODO: straight line homotopy

4. **TODO:**

□

Remark 2.4. *Note that $\star$ operation on the set of path-homotopy equivalence classes of paths in the space X forms a groupoid structure.*

3 The Fundamental Group

Definition 3.1. *Let X be a space and $x_0 \in X$ be a point in X we say that a path $f : I \to X$ is a loop based at x_0 if $f(0) = f(1) = x_0$*

Definition 3.2. *The set of all path-homotopy equivalence classes of loops based at x_0 forms a group under $\star$, called the fundamental group of x based at x_0, denoted as $\pi_1(X, x_0)$*

- *Identity:* $[e_{x_0}]$
- *Inverses:* $[f] \to [\overline{f}]$

Example: $\pi_1(\mathbb{R}^n, (0,0)) \equiv 0 \equiv \langle e_{origin} \rangle$

Proof. Take any two loops $f(t), g(t)$ based at the origin. Then consider the straight-line homotopy

$$H(s,t) = f(s)(1-t) + tg(s)$$

(Check that it is a path homotopy) □

Definition 3.3. *Let α be a path from x_0 to x_1, where $x_0, x_1 \in X$. We define a map $\hat{\alpha} : \pi_1(X, x_0) \to \pi_1(X, x_1)$ such that*

$$\hat{\alpha}([f]) := [\overline{\alpha}] \star [f] \star [\alpha] = [\overline{\alpha} \star f \star \alpha]$$

Proposition 3.4. *$\hat{\alpha}$ is well-defined and an isomorphism.*

Proof. Let $f \simeq_p f'$. We need to check that

$$\begin{aligned} &[\overline{\alpha}] \star [f] \star [\alpha] = [\overline{\alpha}] \star [f'] \star [\alpha] \\ \iff &[\overline{\alpha} \star f \star \alpha] = [\overline{\alpha} \star f \star \alpha] \\ \iff &\overline{\alpha} \star f \star \alpha \simeq_p \overline{\alpha} \star f' \star \alpha \end{aligned}$$

Since $\star$ is well-defined, $[f] \star [\alpha] = [f'] \star [\alpha]$, $\hat{\alpha}$ is well-defined

Let $[f], [g] \in \pi_1(X, x_0)$.

$$\begin{aligned} \hat{\alpha}([f]) \star \hat{\alpha}([g]) &= [\overline{\alpha}] \star [f] \star [\alpha] \star [\overline{\alpha}] \star [f] \star [\alpha] \\ &= [\overline{\alpha}] \star [f] \star [e_{x_0}] \star [g] \star [\alpha] \\ &= [\overline{\alpha}] \star [f] \star [g] \star [\alpha] \\ &= \hat{\alpha}([f] \star [g]) \end{aligned}$$

Hence $\hat{\alpha}$ is a homomorphism. To check that $\hat{\alpha}$ is an isomorphism, we see that $\hat{\hat{\alpha}}(\hat{\alpha})([f]) = [f]$ for all $[f] \in \pi_1(X, x_0)$. □

Corollary 3.4.1. *The fundamental group of X based at x_0 only depends on the path-connected component of $x_0 \in X$ upto isomorphism.*

Definition 3.5. *Let $h : X \to Y$ be a continuous map such that $h(x_0) = y_0$, $x_0 \in X, y_0 \in Y$. Then define $h_\star : \pi_1(X, x_0) \to \pi_1(Y, y_0)$ by $h_\star([f]) = [h \circ f]$. This is a push-forward map.*

Proposition 3.6. *$h_\star$ is a well-defined homomorphism.*

Proof. Let $f \simeq_p f'$. We need to check that $h \circ f \simeq_p h \circ f'$. Let $H : I \times I \to X$ be the path homotopy between f and f'. Consider $h \circ H : I \times I \to Y$.

$$\begin{aligned} h \circ H(s, 0) &= h \circ f \\ h \circ H(s, 1) &= h \circ f' \\ h \circ H(0, t) &= h \circ f(0) \\ h \circ H(1, t) &= h \circ f(1) = h \circ f(0) \end{aligned}$$

Hence $h \circ H$ is the required path-homotopy.

Consider $[f], [g] \in \pi_1(X, x_0)$. We want to show that $h_\star([f]) \star h_\star([g]) = h_\star([f] \star [g])$.

$$h_\star([f]) \star h_\star([g]) = [h \circ f] \star [h \circ g]$$

Note that it suffices to show that $h(f \star g) = (h \circ f) \star (h \circ g)$. Now,

$$\begin{aligned} h(f \star g)(s) &= \begin{cases} (h \circ f)(2s), & s \in \left[0, \frac{1}{2}\right] \\ (h \circ g)(2s-1), & s \in \left[\frac{1}{2}, 1\right] \end{cases} \\ &= (h \circ f) \star (h \star g) \end{aligned}$$

Hence proved. □

Theorem 3.7. *If* $h : (X, x_0) \to (Y, y_0)$ *and* $k : (Y, y_0) \to (Z, z_0)$ *then*

(a) $(k \circ h)_\star = k_\star \circ h_\star$

(b) If id : $(X, x_0) \to (X, x_0)$ *is the identity map then* $(\mathrm{id})_\star$ *is the identity* $\pi_1(X, x_0) \to \pi_((X, x_0)$.

Proof. (a)

$$\begin{aligned} (k \circ h)_\star([f]) &= [(k \circ h) \circ (f)] \\ &= [k(h(f))] \\ &= [k_\star(h_\star[f])] \\ &= (k_\star \circ h_\star)[f] \end{aligned}$$

(b) $(\mathrm{id})_\star[f] = [\mathrm{id} \circ f] = [f]$

□

Corollary 3.7.1. *If* $h : X \to Y$ *is a homeomorphism then* $\pi(X, x_0) \cong \pi(Y, h(x_0)$ *via* $h_\star$.

Definition 3.8. *Let* $P : E \to B$ *be a surjective map. A open set* $U \subseteq B$ *is said to be evenly covered by* P *if* $P^{-1}(U) = \bigsqcup_\alpha V_\alpha$, *where* V_α *are open sets of* E *for all* α *and* $P|_{V_\alpha}$ *is a homeomorphism of* V_α *onto* U.

Definition 3.9. *Let* $P : E \to B$ *be surjective if every point* $b \in B$ *has a neighbourhood* $U \subset B$ *such that* U *is evenly covered by* P. *Then we say that* P *is a covering map and* E *is a covering space of* B.

Examples:

1. id : $X \to X$.
2. $X \times \{1, 2, \cdots, n\} \xrightarrow{P} X$ *where* $P(X, i) = x$.

Theorem 3.10. *The map* $P : \mathbb{R} \to S'$ *such that* $x \mapsto (\cos(2\pi x), \sin(2\pi x))$ *is a covereing space.*

Proof. $P^{-1}(U) = \bigsqcup_{n \in \mathbb{Z}} \underbrace{\left(n - \tfrac{1}{4}\right)\left(n + \tfrac{1}{4}\right)}_{=: V_n}$

Restricted to each V_n the map P is a homeomorphism because $\sin(2\pi x)$ is strictly monotonic on V_n for all n.

TODO: finish $\square$